

Matrix Multiplication

Do transformations ko ek saath combine karne ka tarika

01 Opening Hook

STORY TIME

Socho tumhare paas do machines hain – ek seb ko juice banati hai, doosri juice ko cold karti hai. Agar tum dono machines ko ek line mein laga do, seb ek saath juice bhi banega aur cold bhi hoga. Matrix multiplication bhi bilkul yehi karta hai – **do transformations ko ek mein combine kar deta hai.**

02 Why Does This Concept Exist?

THE PROBLEM

AI models mein data ek layer se doosri layer, phir teesri layer se guzarta hai – har layer apni transformation apply karti hai. Agar hume yeh saari transformations efficiently combine/apply karni hain, toh matrix multiplication hi woh tarika hai jisse hum multiple linear transformations ko systematically chain kar sakte hain.

03 Explain Like I'm 10

SIMPLE WORDS

Socho ek factory hai jisme kaccha maal (raw material) 3 machines se guzarta hai – cutting, painting, packing. Har machine apna kaam karti hai aur output agli machine ko deti hai. Matrix multiplication mein bhi, ek matrix doosre matrix par 'apply' hota hai, aur result teesre matrix ko diya ja sakta hai.

Jab tum ek vector ko matrix se multiply karte ho, har output number us vector ke saare inputs ka weighted sum hota hai – jaise ek voting system jahan har input ka apna vote-weight hota hai.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Matrix multiplication ka core idea hai: row ke elements ko column ke elements se multiply karke add karo (dot product). Yeh operation multiple 'combinations' ko ek saath calculate karta hai - neural network mein yehi step hai jo input features ko naye, useful features mein combine karta hai.

05 Ab Thoda Maths

FORMULA TIME

Rule:

A ($m \times n$) aur B ($n \times p$) ko multiply kar sakte ho sirf jab A ke columns = B ke rows. Result C hoga ($m \times p$).

Formula:

$C[i][j] = \text{sum over } k \text{ of } A[i][k] * B[k][j]$ - matlab har result cell A ki row aur B ke column ka dot product hai.

06 Visualize Everything

PICTURE IT

$$\begin{array}{c|c} A \ (2 \times 2) \\ \hline 1 & 2 \\ \hline 3 & 4 \end{array} \quad \times \quad \begin{array}{c|c} B \ (2 \times 2) \\ \hline 5 & 6 \\ \hline 7 & 8 \end{array} \quad = \quad \begin{array}{c|c|c|c} C = A \times B \\ \hline 1*5+2*7 & 1*6+2*8 & | & 19 \ 22 \\ \hline 3*5+4*7 & 3*6+4*8 & | & 43 \ 50 \end{array}$$

07 AI Connection

WHERE AI USES THIS

Forward Pass: Neural network ka forward pass essentially ek series of matrix multiplications hai: input x weight matrices, layer by layer.

Attention Mechanism: Transformers mein Query, Key, Value matrices ke beech multiplication hi attention scores calculate karta hai.

08 Real Industry Example

REAL COMPANIES

NVIDIA: Tensor Cores khaas taur par matrix multiplication (matmul) operations ko hardware level par accelerate karte hain.

Google DeepMind: AlphaFold jaise models mein bhi lakhon matrix multiplications hoti hain protein structures predict karne ke liye.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch mein matrix multiplication `torch.matmul(A, B)` ya simply `A @ B` se hoti hai. Neural network ki har `nn.Linear` layer internally `input @ weight.T + bias` calculate karti hai forward pass mein.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Matrix multiplication commutative hai ($A \times B = B \times A$).

Sahi baat: Generally $A \times B \neq B \times A$ - order matter karta hai matrix multiplication mein, unlike normal number multiplication.

Galat soch: Koi bhi do matrices multiply ho sakti hain.

Sahi baat: Sirf tabhi ho sakti hain jab pehli matrix ke columns = doosri matrix ke rows.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Matrix multiplication ka basic rule kya hai?

A: Pehli matrix ke columns doosri matrix ke rows ke barabar hone chahiye; result ki shape (pehli ki rows, doosri ki columns) hoti hai.

[Intermediate] Q: Neural network forward pass mein matrix multiplication ka role kya hai?

A: Har layer input vector ko weight matrix se multiply karti hai taaki naye, transformed features nikal sake – yehi 'learning representation' ka core hai.

[Advanced] Q: Batch matrix multiplication kya hoti hai aur AI training mein kyun zaroori hai?

A: Batch matmul multiple samples ko ek saath process karti hai (3D tensors), jisse training GPU par parallel aur fast hoti hai instead of ek-ek sample process karne ke.

12 Exercises

PRACTICE TIME

[Easy] Ek 3x2 matrix ko 2x4 matrix se multiply karo toh result ka shape kya hoga?

Solution: 3x4.

[Medium] $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (identity) ko $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ se multiply karo.

Solution: Result = B khud, kyunki identity matrix koi change nahi karta: $\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$.

[Hard] Kya $A \times B$ hamesha $B \times A$ ke barabar hota hai? Example do jahan yeh false ho.

Solution: Nahi. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. $A \times B$ aur $B \times A$ alag results dete hain – matrix multiplication non-commutative hai.

13 Golden Summary

Concept	Easy Meaning
Matrix Multiplication	Do transformations ko combine karne ka operation
Dot Product Rule	Row x Column, elements multiply karke sum
Shape Rule	$(m \times n) \times (n \times p) = (m \times p)$
Non-commutative	$A \times B$ generally $\neq B \times A$

14 One-Line Memory Trick

“Matrix multiplication = do transformations ko ek machine mein jodna.”